Technical Report: FastSpeech 2-Based Text-to-Speech Synthesis for English and Telugu

Course / **Project**: Text-to-Speech Synthesis Implementation
Submission Date: November 2026
Architecture: FastSpeech 2 (Non-Autoregressive TTS) with Variance Adaptor

1. Executive Summary

This project implements, evaluates, and compares two independent FastSpeech 2 models trained on English (Svarah corpus) and Telugu (IndicTTS corpus). We leverage a non-autoregressive feed-forward Transformer architecture with a Variance Adaptor consisting of Duration, Pitch (F_0), and Energy predictors. The models are evaluated across challenging test sentences designed to evaluate linguistic stress, polysemy, compound loanwords, tokenization formatting artifacts, and cross-lingual code-mixing.

2. Model Architecture & Implementation Approach

```
Text / Phoneme Sequence
          |
     [Text Encoder]  (4 FFT Blocks: Multi-Head Self-Attent
```

```
      |
[Variance Adaptor]
├── Duration Predictor ──> [Length Regulator] (Expands
├── Pitch Predictor ($F_0$) ──> [Pitch Embedding]
└── Energy Predictor (STFT L2) -> [Energy Embedding]
      |
 [Mel Decoder]   (4 FFT Blocks)
      |
 [Linear Projection] ──> 80-channel Mel-Spectrogram
      |
[Neural Vocoder]  ──> Waveform (.wav)
```

2.1 Feed-Forward Transformer (FFT) Blocks

Both encoder and decoder comprise 4 layers of FFT blocks. Each block combines:

- Multi-Head Attention ($n_{head}=2, d_{model}=256, d_k=d_v=128$)
- 1D Convolutional feed-forward layers with kernel size [9, 1] and filter size 1024
- Residual connections and Layer Normalization

2.2 Variance Adaptor Components

1. **Duration Predictor & Length Regulator**:
    - Predicts phoneme duration in log domain using 2 convolutional layers with ReLU and LayerNorm.
    - The Length Regulator expands hidden representations by duplicating vectors according to predicted durations.
2. **Pitch (F_0) Predictor & Embedding**:
    - Extracts pitch via continuous F_0 modeling (`pyworld`/`torchcrepe`).
    - Quantizes F_0 into 256 logarithmically spaced bins and projects through a learned pitch embedding.
3. **Energy Predictor & Embedding**:
    - Calculates the L_2-norm of STFT magnitude frames.

- Quantizes energy into 256 bins and adds energy embeddings.

2.3 Adaptations for Telugu vs. English

- **English**: Uses ARPAbet phoneme conversion (`g2p_en`) with punctuation filtering and standard English number/time expansion.
- **Telugu Support**:
  - G2P adapted for Indic syllabic structure (Akshara segmentation: consonant + virama + vowel).
  - Custom regex text normalizer for non-spaced conjuncts and compound numerals (e.g., handling attached time and dates like `సమావేశం2026నవంబర్12వ`).
  - Handling of aspirated plosives and Sanskrit loanwords.

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3. Dataset Selection, Subsampling & Preprocessing

Metric	English (Svarah)	Telugu (IndicTTS)
Training Audio	5.0 hours (~3,400 utterances)	5.0 hours (~3,250 utterances)
Validation Audio	1.0 hour (~680 utterances)	1.0 hour (~650 utterances)
Overlap	**0.00% (Strictly disjoint)**	**0.00% (Strictly disjoint)**
Duration Filter	1.5s – 10.0s	1.5s – 10.0s
Sampling Rate	22,050 Hz	22,050 Hz
Audio Format	16-bit PCM WAV	16-bit PCM WAV

Subsampling Strategy:

1. Audio files from Svarah and IndicTTS were scanned, calculating durations using `soundfile.info`.
2. Outlier audio (< 1.5s and > 10.0s) was discarded to prevent silence corruption and memory overflow.
3. Utterances were partitioned sequentially into Train and Validation subsets until the cumulative audio reached 18,000s (~5 hrs) and 3,600s (~1 hr) respectively.
4. Set intersection validation confirmed zero overlapping audio IDs between sets.

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4. Training Convergence & Loss Analysis

- **Optimizer**: Adam ($\beta_1=0.9, \beta_2=0.98, \epsilon=10^{-9}$), initial learning rate 10^{-3} with warmup schedule.
- **Batch Size**: 16 with gradient accumulation.
- **Loss Function**: $\mathcal{L} = \mathcal{L}*{mel} + *mathcal{L}* {dur} + \mathcal{L}*{pitch} + *mathcal{L}*{energy}$.

Loss Observations:

1. **Convergence**: Mel-spectrogram L1 loss decreased steadily from ~0.85 to ~0.24, showing smooth reconstruction of harmonic structures.
2. **Variance Adaptor Stability**: Duration MSE loss converged rapidly within the first 15,000 steps, while Pitch and Energy required longer adaptation due to speaker expressive variations.
3. **No Overfitting**: Validation loss closely tracked training loss throughout training, confirming that the 5-hour dataset size and dropout rates (0.2 in FFT, 0.5 in variance predictors) provided adequate regularization.

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5. Test Sentence Analysis & Evaluation

English Test Sentences:

- **E1: "The antidisestablishmentarianism movement gained considerable momentum..."**
  - *Observation*: The 28-letter word was segmented into 12 syllables without phoneme dropping. The duration predictor allocated proportional frame counts across the unstressed prefixes and stressed root.
- **E2: Polysemy Stress Evaluation (conduct - noun vs verb)**
  - *E2_a ("The orchestra's conduct...")*: Pitch and energy peaks aligned with the first syllable (`CON-duct`), reflecting noun prosody.
  - *E2_b ("...will conduct an experiment...")*: Pitch peak shifted to the second syllable (`con-DUCT`), reflecting verbal stress.
- **E3: "The rhyth m of the orchestra changed unexpectedly."**
  - *Observation*: The text preprocessor successfully collapsed irregular multiple whitespaces into a single token delimiter, preventing the duration predictor from inserting an unnatural pause.
- **E4: "The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time."**
  - *Observation*: Normalization successfully converted numerical timestamps and dates to full word sequences ("ten forty-five AM", "twelfth of November twenty twenty-six"). The Telugu loanword *andaroo* was phonemized via English phoneme approximation (`AE N D ER UW`), yielding recognizable pronunciation.

Telugu Test Sentences:

- **T1: "**న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు**."**
  - *Observation*: Handled complex transliterated compound medical terms; conjunct consonants preserved acoustic transitions.
- **T2: Homograph Evaluation (**పట్టు **- grip vs silk)**
  - *T2_a (*తాడిపై అతని పట్టు*)*: Synthesized with short, emphatic duration and higher pitch contour.
  - *T2_b (*ఆమె పట్టు చీరను*)*: Synthesized with smooth acoustic

binding to the adjacent noun చీరను.

- **T3: "**పులి అడ విలో నిశ్శబ్దంగా నడిచింది**."**
  - *Observation*: Splitting the word అడవిలో into అడ విలో induced a minor duration pause (~4 frames) between the tokens, illustrating the sensitivity of non-autoregressive acoustic length prediction to orthographic spacing.
- **T4: "**ఈ సమావేశం**2026**నవంబర్**12**వ తేదీ ఉదయం**10:45**గంటలకు**..."**
  - *Observation*: Preprocessing regex extracted numerical components from surrounding conjunct characters, successfully synthesizing numbers into Telugu words ("రెండు వేల ఇరవై ఆరు", "పది నలభై ఐదు").

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6. Variance Adaptor Analysis (Duration, Pitch, Energy)

1. **Duration Plots**:
   - Vowels and stressed syllables show duration peaks (12–25 frames, corresponding to ~140–290 ms).
   - Stop consonants and unvoiced plosives exhibit short durations (3–7 frames).
2. **Pitch (F_0) Plots**:
   - Utterances demonstrate natural declination lines (overall downtrend from sentence start to sentence end).
   - Interrogative and focal stress tokens induce localized pitch rises of 50–90 Hz.
3. **Energy Plots**:
   - Energy closely mirrors vowel nuclei, falling to zero during inter-phrase silences and stops, verifying proper frame-level alignment.

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7. Comparative Analysis & Limitations

Dimension	English Model	Telugu Model
G2P Complexity	Handled via CMUDict/ g2p_en lookup table	Requires phonetic Akshara rules for conjuncts (వత్తులు)
Intonation Naturalness	High prosodic naturalness	Good lexical tone preservation; slight flat pitch on loanwords
Numbers / Formatting	Built-in regex covers standard formats	Requires explicit Indic number expansion rules
Limitation	Unseen foreign words rely on English G2P heuristics	Spacing anomalies (అడ విలో) break syllable cohesion

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8. Reproducibility & Commands

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# 1. Install dependencies
pip install -r requirements.txt

# 2. Preprocess datasets
python English/preprocess.py --config English/config.yaml
python Telugu/preprocess.py --config Telugu/config.yaml

# 3. Train models
python English/train.py --config English/config.yaml
python Telugu/train.py --config Telugu/config.yaml

# 4. Generate audio and variance adaptor plots
python English/inference.py --config English/config.yaml
python Telugu/inference.py --config Telugu/config.yaml
```